

Nurkanat Baisenkul

Platform, Solutions & Security Engineer

Almaty, Kazakhstan

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Platform- and security-focused engineer with a cybersecurity background and hands-on DevOps experience hardening enterprise infrastructure. Builds secure delivery platforms — CI/CD with integrated SAST, secrets management with HashiCorp Vault and PKI, and highly available data and messaging services — and ships open-source security tooling in Rust, C, and Python. Comfortable across cloud (AWS), infrastructure-as-code, and applied AI — including LLM agents grounded in knowledge graphs, the subject of published IEEE research.

SKILLS

Cloud & IaC	AWS, Terraform, AWS CloudFormation
Platform / CI-CD	Docker, Kubernetes, GitLab CI, Jenkins, GitHub Actions, on-prem GitLab & runners
Security	HashiCorp Vault, PKI / SSL, SAST, network security, password-policy enforcement
Languages	Python, Rust, C, Bash, SQL
Data & Infra	PostgreSQL (replication), RabbitMQ (clustering), BIND9 DNS, Linux administration
AI	LLM agents, retrieval-augmented generation, knowledge graphs (RDF / SPARQL), applied ML with Python

EXPERIENCE

Kaspi.kz

May 2025 – Present

Cyber Defense Process Architect

Almaty, Kazakhstan

- Review the security architecture and deployment approach of services across hundreds of engineering teams, catching misconfigurations and low-hanging risks before sign-off and hand-off to the Ethical Hacking team.
- Led on-premise proof-of-concept pilots of Kubernetes security platforms — Luntry and Aqua Security — largely single-handedly: deploying each, exercising its full feature set, and assessing operational fit and trade-offs for the bank.
- Produce security architecture diagrams and evaluate deployment designs for containerized (Kubernetes) workloads.
- Verify network configuration — NAT, VLANs, routing, and IPSec tunnels — on Huawei firewalls during architecture reviews.

Institute of Space Engineering and Technologies LLP

Feb 2025 – Aug 2025

DevOps Engineer (Junior → Middle)

On-site → Remote (part-time)

- Built the organization's CI/CD platform from the ground up, single-handedly — on-premise GitLab CE with Docker-in-Docker runners serving 20+ component repositories, with 4–5 reusable pipeline templates covering the different languages and build approaches in use; each job ran in its own dependency-prebaked container to cut build times and keep the runner host clean.
- Engineered per-project deployment stages that ran developer-supplied config scripts, executed checks, and restarted services — and built a custom approval gate (absent from GitLab CE) that verified the triggering user against an allowlist and blocked release until a manager manually approved.
- Configured PostgreSQL replication and logging (2–3 node cluster) and a high-availability RabbitMQ cluster (2–3 nodes); authored a custom C passwordcheck module enforcing organizational password policies.
- Deployed HashiCorp Vault for secrets management — path policies, roles, and SSL certificate issuance via Vault PKI; administered BIND9 DNS and hardened Linux hosts (users/groups/permissions, secure SSH/FTP, port and cron management).

EPAM Kazakhstan

Jul 2024 – Dec 2024

Cloud & DevOps Trainee

Almaty, Kazakhstan

- Built and operated containerized workloads with Docker and Kubernetes across Jenkins, GitLab, and GitHub CI/CD pipelines.
- Deployed applications to AWS and provisioned infrastructure as code with CloudFormation and Terraform.

Development Bank of Kazakhstan

Mar 2023 – May 2023

Cybersecurity Engineer (Intern)

Astana, Kazakhstan

- Conducted network security testing in Python and Rust alongside the bank’s security team.
- Built an ARP-spoofing detection tool in Rust — packaged as a Windows service — that monitors the LAN and forwards alerts to a Syslog server; deployed on machines across the corporate network.

PROJECTS

financial-kg-agent (Python)  <https://github.com/tuchaVshortah/financial-kg-agent>
LLM agent that grounds financial-transaction queries in a knowledge graph — builds an RDF graph, retrieves facts over SPARQL, and injects them into the model with compliance guardrails against reasoning from incomplete data. Implementation behind the published research below.

passwordcheck (C)  <https://github.com/tuchaVshortah/passwordcheck>
PostgreSQL passwordcheck implementation that enforces configurable password policies.

arp-spoofing-detector-rs (Rust)  <https://github.com/tuchaVshortah/arp-spoofing-detector-rs>
Detects ARP-spoofing attacks on local or corporate networks, runs as a Windows service, and ships logs to a Syslog-compatible server.

puff (Python)  <https://github.com/tuchaVshortah/puff>
Passive subdomain enumeration tool for reconnaissance.

EDUCATION

Suleyman Demirel University Sep 2024 – Jun 2026
Master of Computer Science *Graduated*

- Focus: Python, Artificial Intelligence, programming.

Astana IT University Sep 2021 – Jun 2024
Bachelor of Cybersecurity

- Focus: web development, cybersecurity, network security.

PUBLICATIONS

Investigating Knowledge Graphs for Context-Aware Search in Financial Transactional AI Agents 2026
ICECCO 2026 — indexed in IEEE Xplore
 orcid.org/0009-0000-8386-4960

LANGUAGES

English — Fluent (Duolingo English Test 135/160, CEFR C1) • **Kazakh** — Native • **Russian** — Fluent